SDG BLOCKCHAIN ACCELERATOR

Impact Measurement Framework

1. Project Information

- **Project Name:** SHELTER
- **Challenge & UNDP Office:** MALAWI UNDP
- **Document Version:** 1.0

2. SDG Alignment

- **SDG 09: Industry, Innovation and Infrastructure**
  - Target**:** Significantly increase access to information and communications technology and strive to provide universal and affordable access to the Internet in least developed countries by 2030.
  - Indicator**:** Number of individuals reached who utilize Basic Feature Phones (non-smartphones) to access financial aid services via SMS infrastructure.

- **SDG 10: Reduce Inequalities**
  - Target: By 2030, empower and promote the social, economic and political inclusion of all, irrespective of age, sex, disability, race, ethnicity, origin, religion, economic or other status.
  - Indicator: Proportion of people living below 50 percent of median income, by age, sex, and persons with disabilities.

3. Key Performance Indicators (KPIs)

KPI	Measurement Method	Target / Threshold	Notes
Number of successful validator transactions	Blockfrost API / Cardano Explorer logs	95%+ success rate	
Active users onboarded	LTW Service	200 users	Tracked via wallet addresses
Number of people reached who do not have access to formal financial products or services	LTW Service	200 users	Directly measures financial inclusion enabled by the SMS Wallet infrastructure.

Proportion of people living below 50% of median income, by age, sex, and persons with disabilities	Demographic breakdown of the "Beneficiary Upload" CSV files provided by the NGO.	100% of target group	Ensures funds are strictly routed to the identified vulnerable populations verified by the NGO.

4. Tracking Methods

- **On-chain Metrics:**
    - **Transaction success rate:** Verified via **Blockfrost API** queries filtering by script address and transaction status.
    - **Total Volume Disbursed:** Sum of ADA values in UTxOs consumed by BeneficiaryAction transactions.
    - **Wallet Activation:** Number of distinct staking keys interacting with the SHELTER protocol.
- **Off-chain Metrics:**
    - **Information provided by the NGO:** Demographic data parsed from bulk upload CSVs (Age, Sex, Location, Disabilities, Vulnerability Status).
    - **Field Data:** Surveys or audits conducted by field agents during the onboarding process.

5. Baseline vs Projected Outcomes

Metric	Baseline	Projected Outcome
Active users	0	200
Transaction success rate	N/A	≥95%
Amount Disbursed (ADA)	0	≥5000
Avg. Time to Receive Aid	3-5 Days (Traditional Cash)	< 10 Minutes (SHELTER)

6. Monitoring & Reporting Plan

The monitoring strategy employs a Dual-Layer Verification approach. We cross-reference off-chain user data from the LTW Service (Backend) with immutable on-chain records from the Cardano Blockchain to ensure 100% data integrity and transparency.

- ***Frequency:*** *Quarterly Impact Reports*
- ***Responsible Party:*** *Project Manager*
- ***Format:***
    - ***Technical:*** *On-chain execution logs (Blockfrost) compared against backend activity logs.*
    - ***Impact****: PDF/CSV Summaries generated via* ***direct on-chain queries*** *visualizing funds disbursed vs. claimed.*
    - ***Off-chain****: information provided by the NGO working on the field implementation with end-beneficiaries.*
- ***Tools:***
    - ***LTW Backend Service:*** *Primary source for user activity logs, SMS history, and beneficiary metadata.*
    - ***Blockfrost API:*** *Used to query the Cardano Ledger for independent verification of transaction finality and UTxO status.*
    - ***LTW Database:*** *Primary Administrative Interface. Currently used via direct query/CLI to monitor user onboarding, system status, and generate data exports for the NGO.*
    - *NGO´s Surveys: end-beneficiaries information provided to the NGOs working on the field.*

7. Risks & Mitigation

- **Risk**: Partial or Delayed information from the entity disbursing funds.
    - **Mitigation**: Impact dashboard with the information to do bulk creation and transfer
- **Risk**: Multi-party organizations taking part of the disbursement and fractional information.
    - **Mitigation**: Implementation of the **Rahat Dashboard** standards to aggregate information from multiple participants into a single, interoperable source of truth compatible with global humanitarian metrics.
- **Risk**: Incomplete surveys from the people in the field.
    - **Mitigation**: Listing a form that needs to be completed to be eligible for the support or aid.